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SCHOOL OF ENGINEERING AND TECHNOLOGY

SUMMER TRAINING REPORT

**Solar Power Generation Analysis and Performance Monitoring Using
Python**

Submitted in partial fulfillment of the
Requirements for the award of
Degree of Bachelor of Technology
in
Artificial Intelligence and Data Science

Submitted By

Name: Dhruv Saluja

Enrollment No.: 51317711924

Semester: V

Academic Year: 2026–27

DECLARATION

I, Dhruv Saluja, bearing Enrollment No. 51317711924, a student of B.Tech. in Artificial Intelligence and Data Science (AI&DS) at the School of Engineering and Technology, Vivekananda Institute of Professional Studies – Technical Campus, Pitampura, New Delhi, hereby declare that the Summer Training Report entitled “Solar Power Generation Analysis and Performance Monitoring Using Python” is an authentic record of my own work carried out during the Summer Training undertaken at Kiefer India Private Limited, Rohtak, Haryana, from 15 July 2026 to 15 August 2026, as a partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science.

To the best of my knowledge, the data, analyses, and findings presented in this report are genuine and accurate. This report has not been submitted, either wholly or in part, to any other institution or university for the award of any degree, diploma, or certification.

Dhruv Saluja
51317711924

Date:

Place:

ACKNOWLEDGEMENT

I express my sincere gratitude to **Kiefer India Private Limited**, Rohtak, Haryana, for giving me the opportunity to undertake my summer training from 15 July 2026 to 15 August 2026 and to work on the project "Solar Power Generation Analysis and Performance Monitoring Using Python". The professional environment of the company gave me a real understanding of how data analytics supports the renewable energy sector.

I am deeply thankful to my industry mentor, Mrs. Rashmi Arora, Director of Kiefer India Private Limited, for patient guidance and technical support throughout the project, and for academic supervision that helped align this work with the standards of the university. I also thank, Dr.Sonakshi Vij, Head of the Department of Artificial Intelligence and Data Science, and all faculty members for the foundation in programming, statistics and machine learning that made this work possible.

I am grateful to Vivekananda Institute of Professional Studies – Technical Campus and GGSIP University for including summer training in the curriculum, and finally to my family and friends for their constant encouragement and support.

Submitted By:
DHRUV SALUJA
51317711924

INTERNSHIP OFFER LETTER



13th July 2026

To,
Mr. Dhruv Saluja

SUBJECT: OFFER LETTER FOR TECHNICAL EXECUTIVE INTERNSHIP

Dear Dhruv,

We are pleased to inform you that, following your application and interview, Kiefer India Private Limited has selected you for the position of Technical Executive Intern. Your internship details are set out below:

PARTICULARS	DETAILS
Position	Technical Executive Intern
Internship Duration	1 Month
Mode	Offline
Stipend	Unpaid
Office Timings	9:00 AM – 6:00 PM
Work Location	Mansarovar Colony, Rohtak, Haryana
Start Date	15 July 2026
End Date	15 August 2026

During the internship, you are expected to maintain professionalism, follow company policies, and complete tasks assigned by your reporting manager, gaining practical, hands-on industry exposure. Please report to the above location on your joining date with a valid photo ID and any documents requested by HR.

We believe your enthusiasm and dedication will make you a valuable addition to our team, and we look forward to welcoming you. Congratulations, and best wishes for a successful internship!

Authorized Signatory



Rashmi Arora
Director

Kiefer India Private Limited
Email – kieferindia@gmail.com
+91 9711870401

Regd. office: 86-A, Mansarovar Colony, Rohtak, Haryana 124001, India
Email: kieferindia@gmail.com

INTERNSHIP COMPLETION CERTIFICATE

Certificate No.: KIPL/INT/2026/001



CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that

Dhruv Saluja

has successfully completed his internship as a **Technical Executive Intern**
at Kiefer India Private Limited.

The internship was conducted from 15 July 2026 to 15 August 2026 (1 Month)
in Offline mode at our office located at Mansarovar Colony, Rohtak, Haryana.

During this period, he demonstrated dedication, enthusiasm and a sincere commitment
towards learning and performing the assigned tasks.

We wish him all the best for his future endeavors.



Rashmi Arora

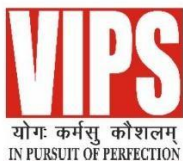
Director
Kiefer India Private Limited

Regd. Office: 86-A, Mansarovar Colony,
Rohtak, Haryana 124001, India

Email: kieferindia@gmail.com

+91 9711870401

EMPLOYER / MENTOR FEEDBACK FORM



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Employer/Mentor Feedback Form

Name of the Employer: Rashmi Arora	Organization: Kiefer India Private Limited
Name of the Employee: Dhruv Saluja	Role of the Employee: Technical Executive Intern

Parameter for Assessment	Highly Satisfied	Satisfied	Moderately Satisfied	Dissatisfied	Highly Dissatisfied
1. Relevant technical knowledge relevant essential to do his/her assignments.	✓				
2. Ability to identify, analyze and research literature for assigned problems.	✓				
3. Ability to exhibit awareness of societal, health, safety, legal and cultural responsibilities relevant to the field.	✓				
4. Ability to exhibit ethical growth and professionalism.	✓				
5. Ability to complete assignments, meet deadlines, understand instructions and work with others.	✓				
6. Oral and written communication skills of the intern	✓				

7. Please suggest the requisite modification in the curriculum/ other feedback to make the students industry ready

Demonstrated good technical understanding and a positive attitude toward learning.

Completed assigned tasks responsibly and worked well with the team.

Further exposure to practical industry tools and projects is recommended.

Date: **14 August 2026**

Signature :



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CHAPTER 1: INTRODUCTION

1.1 Renewable Energy Overview

Global demand for electricity keeps rising as populations grow and developing economies industrialise. For more than a century this demand has been met mainly by burning coal, oil and natural gas. The cost of that dependence is now clear: combustion releases carbon dioxide and methane, which trap heat in the atmosphere and drive global warming, while also causing air pollution, water contamination and habitat loss. Because fossil fuels are finite, a shift towards sustainable energy is both an environmental and an economic necessity.

Renewable energy comes from sources that replenish naturally on a human timescale. The most widely deployed forms are solar, wind, hydropower, biomass and geothermal energy (Figure 1.1). Hydropower and wind are established but depend on suitable geography, biomass needs careful logistics to remain carbon-neutral, and geothermal is viable mainly near tectonic boundaries. Solar energy, in contrast, is available almost everywhere, scales from a rooftop panel to a utility-scale park, and has become far cheaper over the past decade. Solar installations also have no moving parts, which keeps their maintenance needs low.

Each source uses a different natural phenomenon. Solar energy captures the light and heat of the sun and is scalable from a small rooftop panel to a large solar farm. Wind turbines convert the kinetic energy of moving air into mechanical power and then electricity. Hydropower harnesses the gravitational energy of falling or flowing water, usually through dams or run-of-the-river systems. Biomass energy comes from agricultural residues, wood and animal waste, which can be burned or converted into biogas and biofuels. Geothermal energy taps the internal heat of the earth through hot water or steam.

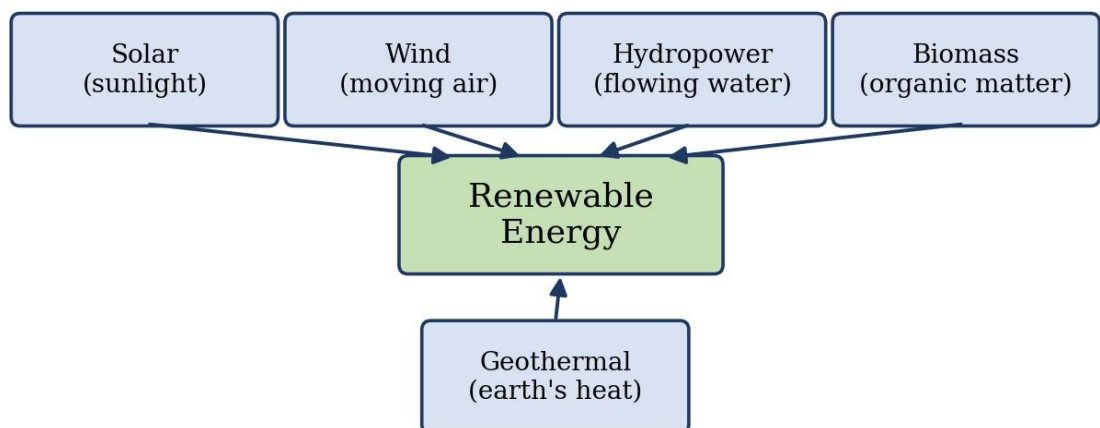


Figure 1.1: Renewable Energy Sources

1.2 Solar Energy and Its Growth in India

Solar energy is the radiant light and heat of the sun. It is converted directly into electricity by photovoltaic (PV) systems, whose solar cells are made of semiconductor material, usually silicon. The electricity a panel produces is direct current (DC), whereas the grid and most appliances use alternating current (AC). An inverter therefore sits between the panels and the grid and converts DC into AC at the required voltage and frequency.

India is well placed for solar power. Most regions receive strong sunshine on roughly 300 days a year, and the country has backed this potential with policy: the Jawaharlal Nehru National Solar Mission set national capacity targets, and India co-founded the International Solar Alliance to promote solar deployment worldwide. Large solar parks have been commissioned in land-rich states such as Rajasthan and Gujarat, while rooftop and agricultural solar systems spread generation to cities and farms. Solar power now plays a central role in India's efforts to meet rising demand, cut emissions and improve energy security.

1.3 Importance of Solar Power Plants

Large solar power plants, or solar parks, feed electricity directly into the transmission grid and can supply cities, industrial complexes and entire districts. Their most important benefit is environmental: unlike thermal plants, they emit no carbon dioxide, sulphur dioxide or nitrogen oxides during operation and produce no ash that contaminates local water tables. By replacing fossil-fuel generation they contribute directly to mitigating climate change and improving regional air quality.

Solar plants also bring economic benefits. Their construction, operation and maintenance create direct and indirect employment for engineers, technicians and local workers, and land leases provide a steady income in the rural areas where parks are usually located. Because the fuel is sunlight, the cost of generation is insulated from fuel-price shocks and supply disruptions and remains predictable over a plant life of 25 to 30 years. For a country that imports coal, oil or gas, every unit of solar electricity also reduces import dependence and strengthens energy security.

1.4 Need for Performance Monitoring

Solar plants are generally reliable and need far less physical maintenance than thermal plants, but they are not maintenance-free. A plant contains thousands of interconnected panels, long cable runs and many high-capacity inverters, and its output changes constantly with the sun's angle, cloud cover, atmospheric dust and air temperature. Unlike a coal plant, whose output can be held steady, a solar plant depends entirely on fluctuating conditions. Because of this

variability, a drop in total output does not by itself reveal whether a passing cloud or a failing component is responsible.

Inverters deserve particular attention. They run continuously under high electrical and thermal stress, and when one underperforms, the DC power from every string connected to it is lost. Panels also degrade gradually through soiling, cracks and ageing. Without a data-driven approach, such gradual losses can go unnoticed for weeks. Systematic monitoring of generation and weather data, as described in the international standard IEC 61724-1 [1] and in reviews of PV monitoring systems [2], lets operators compare actual output with expected output and move from reactive to proactive maintenance.

Panels are constantly exposed to the environment as well. Dust, bird droppings and microscopic cracks steadily reduce the amount of light a panel can absorb, and without data these degradations can go unnoticed for weeks or months. A monitoring system solves this by continuously collecting time-series data from generation meters, inverters and weather sensors, and by establishing a baseline of expected performance. When actual generation departs from the baseline, an early fault can be flagged, allowing operators to service underperforming equipment before large energy losses occur.

1.5 Problem Statement

Modern solar plants log DC power, AC power, inverter status and weather variables every few minutes. The challenge is to turn these raw, noisy records into actionable information. In a plant with many inverters, a fall in total yield is easy to see on the main meter, but locating the exact unit responsible, and separating a genuine fault from the effect of weather, is difficult. Without a systematic method, subtle faults compound until equipment fails and a significant share of potential energy is lost.

This project addresses that gap by building a Python-based analytical workflow that cleans raw sensor logs, calculates inverter efficiency, relates power output to weather variables and applies a clear numerical rule to flag underperforming inverters.

1.6 Objectives of the Project

The project aims to develop a programmatic approach to evaluate solar plant performance using Python. Table 1.1 lists the specific objectives.

Table 1.1: Project Objectives

Objective	Description
Data Cleaning	Handle missing values, correct erroneous sensor entries and standardise date-time formats.
Data Preparation	Merge the generation and weather logs into one dataset on synchronised time stamps.
Power Generation Analysis	Study daily DC and AC power trends, peak periods and overall yield patterns.
Inverter Performance Analysis	Compare the cumulative yield of individual inverters to separate high and low performers.
Efficiency Analysis	Calculate the AC/DC conversion ratio while excluding non-generating night-time periods.
Weather Impact Analysis	Quantify how irradiation, ambient temperature and module temperature affect output.
Fault Detection	Flag inverters whose conversion efficiency falls below 90% during daylight hours.
Performance Monitoring	Combine all findings into a view of plant health that supports proactive maintenance.

1.7 Scope of the Project

The project covers code-driven analysis of a historical solar plant dataset with standard Python libraries: the DC-to-AC relationship, inverter-wise yield comparison, efficiency evaluation, weather impact study and rule-based fault detection. It does not include machine learning or deep learning models, forecasting of future generation or weather, or IoT and real-time systems. All results are derived from static historical data using descriptive statistics and rule-based logic.

1.8 Organization of the Report

The report is organised into eight chapters, summarised in Table 1.2. The first three chapters set the context, the next two provide technical and data background, and the last three describe the method, the results and the conclusions.

Table 1.2: Organization of the Report

Chapter	Content
Chapter 1	Introduction: renewable energy, solar growth in India, need for monitoring, problem statement, objectives and scope.
Chapter 2	Organization and internship overview, work undertaken and skills acquired.
Chapter 3	Literature review, summary of selected studies and research gap.
Chapter 4	Fundamentals of solar power systems, economic aspects and government support.
Chapter	Content

Chapter 5	Dataset description and development environment.
Chapter 6	Methodology: workflow, cleaning, preprocessing and the analytical modules.
Chapter 7	Results and discussion of each analysis.
Chapter 8	Challenges, learning outcomes, conclusion and future scope.

1.9 Chapter Summary

This chapter introduced the need for renewable energy and explained why solar power has grown quickly in India. It described how photovoltaic systems convert sunlight into DC electricity and how inverters supply AC power to the grid. It then argued that variable weather and inevitable hardware ageing make continuous, data-driven performance monitoring essential, which led to the problem statement of locating under-performing equipment in large, noisy datasets. Finally, it set out the eight objectives of the project and its scope, which is limited to descriptive analytics and rule-based logic on historical data.

CHAPTER 2: ORGANIZATION AND INTERNSHIP

OVERVIEW

2.1 About the Organization

Kiefer India Private Limited, based in Rohtak, Haryana, works in the renewable energy analytics sector. The company applies data-driven methods to monitor and optimise energy assets, and it sits at the intersection of energy engineering and data science, turning the large volumes of sensor data produced by modern plants into insights for operators and facility managers. Its work relies on programming languages such as Python and on structured data pipelines and visualisation, which reflects how digital technology has become central to modern industry. Table 2.1 summarises the training details.

Organisations can no longer rely only on manual inspection and reactive maintenance. Modern industry needs intelligent monitoring systems, structured data pipelines and clear visualisations to remain competitive, and companies that offer these capabilities help others to optimise their physical infrastructure.

Table 2.1: Organization and Internship Details

Particular	Details
Organization	Kiefer India Private Limited
Location	Mansarovar Colony, Rohtak, Haryana, India
Position	Technical Executive Intern
Domain	Renewable energy analytics
Training period	15 July 2026 to 15 August 2026 (one month)
Mode of training	Offline (on-site)
Industry mentor	Mrs. Rashmi Arora, Director
Project title	Solar Power Generation Analysis and Performance Monitoring Using Python

2.2 Internship Overview

The internship ran for one month, from 15 July 2026 to 15 August 2026, and was completed on-site at the company's Rohtak facility. Its purpose was to gain hands-on experience in data analytics applied to renewable energy and to bridge the gap between classroom learning and industrial practice. Working in the company gave direct access to experienced analysts and

exposure to how commercial teams structure data pipelines, manage timelines and communicate results.

2.3 Objectives of the Internship

1. **Understand industrial work culture:** observe how technical teams collaborate and meet deliverables.
2. **Apply academic knowledge:** use programming and data-structure concepts from the B.Tech curriculum on a live project.
3. **Learn practical data analysis:** follow the full lifecycle from ingestion and cleaning to visualisation.
4. **Improve programming skills:** gain proficiency in Pandas, NumPy, Matplotlib and Seaborn.
5. **Develop problem-solving ability:** debug code, handle erroneous data and design rules to detect faults.
6. **Understand project documentation:** keep clear records of code and findings for professional reporting.

2.4 Work Undertaken

The core assignment was the project "Solar Power Generation Analysis and Performance Monitoring Using Python", which processed and evaluated historical data from a solar power plant. The first phase was dataset understanding: reviewing the raw CSV files to learn the structure of the generation logs, identifying key features such as DC power, AC power and daily yield, and working out how they relate to the accompanying weather sensor data.

Real sensor data is rarely perfect, so significant effort then went into cleaning and preprocessing. This included standardising inconsistent date-time formats across the files, handling missing values logically and merging the generation and weather datasets into one structured dataset suitable for querying.

Analysis then proceeded module by module. Power generation analysis aggregated the cleaned data to establish daily trends and the plant's baseline capacity. Inverter performance analysis grouped cumulative yield by inverter ID to separate the highest contributors from the lowest. Efficiency analysis calculated the AC/DC ratio, with conditional logic to exclude non-generating night-time intervals that would otherwise distort the metric. Weather impact analysis applied statistical methods to relate temperature and irradiation to electrical output. The final task was a

fault detection rule that flags any inverter falling below 90% conversion efficiency, giving an automated way to identify underperforming hardware.

2.5 Skills Acquired and Academic Relevance

The internship built both technical and professional competencies, summarised in Table 2.2. It also showed how academic ideas such as loops, data structures, correlation and aggregation are used to prevent equipment failure and protect financial returns, something idealised classroom datasets rarely demonstrate.

Table 2.2: Skills Acquired During the Internship

Category	Skill	Application
Technical	Python	Primary language for all scripting and analysis logic.
Technical	Pandas and NumPy	Cleaning, merging and grouping time-series data; numerical calculations.
Technical	Matplotlib and Seaborn	Line charts, scatter plots, bar charts and correlation plots.
Technical	Data analysis	Extracting operational insight from raw industrial metrics.
Professional	Documentation	Commented code and a structured written report.
Professional	Communication	Explaining findings and methods to mentors.
Professional	Problem solving	Debugging errors and handling data anomalies.
Professional	Analytical thinking	Breaking complex datasets into calculable metrics.

2.6 Chapter Summary

This chapter described Kiefer India Private Limited and its work in renewable energy analytics. It gave the details of the one-month, on-site internship from 15 July 2026 to 15 August 2026, listed six internship objectives and summarised the work carried out on the solar monitoring project, from dataset understanding to fault detection. The skills gained, both technical and professional, were tabulated, and the relevance of the internship to the B.Tech curriculum was explained.

CHAPTER 3: LITERATURE REVIEW

3.1 Introduction

A literature review establishes what earlier work has achieved, which methods have proved practical and where gaps remain. This chapter reviews published work on solar plant monitoring, data analytics, inverter performance, weather effects and fault detection, and ends with the research gap that motivates the present project.

3.2 Solar Power Plant Monitoring Systems

Monitoring of PV plants has evolved from manual meter reading to automated data acquisition, typically through SCADA systems that log DC and AC power, voltage, current and environmental variables. Madeti and Singh reviewed monitoring systems in terms of sensors, data-acquisition controllers, data transmission and data storage and analysis, and noted that cost and complexity limit advanced monitoring mainly to large plants [2]. Rahman et al. surveyed modern monitoring systems worldwide [3]. The IEC 61724-1 standard defines how PV system performance should be monitored and reported [1].

A widely used indicator is the performance ratio, which compares the energy actually produced with the energy expected from the plant's rating. Dierauf et al. point out that this ratio depends on weather as well as on system quality, and they propose a weather-corrected version so that performance can be compared fairly across seasons [4]. The idea that output must be interpreted against weather underpins the weather impact module of this project.

Monitoring studies typically follow a common pattern: data is extracted from an operating plant, noisy sensor readings are cleaned, and high-frequency records are aggregated into hourly, daily or monthly values so that macroscopic trends become visible. The practical value of such monitoring is that operators can see quickly when actual generation departs from the expected baseline, and can therefore detect equipment problems earlier than they would through manual inspection. This is the methodological basis for the data preparation and power generation modules of the present work.

3.3 Data Analytics in Renewable Energy

A PV plant is a continuous source of time-series data, and descriptive analytics such as grouping by time period or by equipment identifier is often enough to expose under-performing sections. Reviews of PV reliability show that performance and degradation analysis are central to long-

term plant management [5], and that module degradation is typically gradual rather than sudden [6]. Gradual losses of this kind are hard to notice without trend analysis. Economic analysis of maintenance strategies also shows that the choice between reactive and planned maintenance has a direct financial effect on plant operation [7]. Reliable analytics depends on careful preprocessing, because sensor logs contain missing time stamps, null values and inconsistent formats.

Programming languages such as Python and R are widely used for this work because they handle large-scale data manipulation efficiently. A useful consequence is that plant managers can be given clear visual summaries, such as daily yield trend lines or bar charts comparing inverters, and can direct maintenance crews to a specific array connected to a specific inverter instead of asking them to inspect thousands of panels. Targeted maintenance of this kind reduces cost and downtime.

3.4 Inverter Performance and Efficiency

The inverter is the point through which all generated power must pass, so its behaviour strongly affects plant output. Its conversion efficiency is expressed as the ratio of AC output to DC input, and identical inverters at one site can be compared through their accumulated yields. Field studies of grid-connected PV systems, such as those in Singapore [8] and Thailand [9], analyse monitored data to derive performance indicators of this kind. An inverter whose efficiency drifts downwards, or whose yield lags behind its peers, is a candidate for inspection.

Modern inverters are highly efficient, with conversion efficiencies that can exceed 98% under good conditions, but real-world efficiency varies with temperature, load and age. Continuous monitoring of the conversion ratio is therefore one of the most reliable ways of detecting early hardware degradation, because a failing inverter shows a slowly falling efficiency long before it stops working altogether. Comparing the yield of inverters at the same site adds a second, independent view, since yield differences that weather cannot explain point to localised problems.

3.5 Weather Impact on Solar Power Generation

Irradiation is the main driver of PV output, and DC power rises almost linearly with it. Temperature acts in the opposite direction: Skoplaki and Palyvos compiled the correlations showing that the efficiency of silicon modules falls as operating temperature rises [10]. A robust monitoring approach must therefore include weather data, so that a hot or cloudy day is not mistaken for hardware failure.

Studies of weather effects usually collect synchronised generation and meteorological data and use correlation analysis and scatter plots to judge the strength and direction of each relationship. The result is that the highest yields tend to occur on clear, cool days with high irradiation rather than at the peak of summer heat, and that a hot, cloudy day should not be reported as an equipment failure. Weather impact analysis is therefore an essential part of any reliable monitoring approach.

3.6 Fault Detection in Solar Power Systems

Fault detection has been reviewed extensively [11], [12], [13], [14]. Approaches range from comparing measured behaviour with model-predicted behaviour, as in the method of Chine et al. for grid-connected plants [15], to machine-learning classifiers. Simple rule-based methods that compare a performance indicator with a fixed threshold are computationally light, transparent to maintenance staff and require no labelled training data, which makes them attractive where such data is unavailable. The present project follows this rule-based approach.

Faults range from complete failures, such as a blown fuse or disconnected wire, to gradual degradation, such as dust accumulation on panels or ageing components inside an inverter. A rule of the form "flag the inverter if its efficiency stays below a chosen percentage" can catch the second kind at an early stage. Detecting an inverter that has slipped to 89% efficiency before it fails completely allows repairs to be scheduled outside peak hours, which preserves energy yield and reduces cost.

3.7 Summary of Selected Studies

Table 3.1 summarises the studies most relevant to this work and how each informs the project.

Table 3.1: Summary of Selected Studies

Ref.	Authors (Year)	Focus	Relevance to this project
[2]	Madeti and Singh (2017)	Review of PV monitoring systems	Framework for sensors, data acquisition and analysis.
[3]	Rahman et al. (2018)	Global modern monitoring systems	Context for data-driven plant monitoring.
[5]	Sharma and Chandel (2013)	Performance and degradation analysis	Basis for long-term performance evaluation.
[6]	Jordan and Kurtz (2013)	Analytical review of PV degradation rates	Shows degradation is gradual, motivating trend analysis.
[10]	Skoplaki and Palyvos (2009)	Temperature dependence of module efficiency	Basis for the temperature impact analysis.
[4]	Dierauf et al. (2013)	Weather-corrected performance ratio	Supports interpreting output against weather.

Ref.	Authors (Year)	Focus	Relevance to this project
[15]	Chine et al. (2014)	Fault detection in grid-connected plants	Model-based alternative to rule-based detection.
[11]	Madeti and Singh (2017)	Types of PV faults and detection techniques	Overview of fault categories and methods.
[12]	Mellit et al. (2018)	Fault detection and diagnosis methods	Comparison of detection approaches.
[13]	Triki-Lahiani et al. (2018)	Fault detection and monitoring systems	Links monitoring techniques to failure types.

The studies fall into four groups. The monitoring reviews [2], [3] and the IEC standard [1] describe what should be measured and how it should be stored. The reliability and degradation studies [5], [6] show that losses build up slowly, which favours trend-based analysis. The work on temperature and weather correction [10], [4] explains why output has to be read against conditions. The fault-detection reviews and methods [11], [12], [13], [14], [15] cover the range from simple rules to model-based and learning-based detection.

Taken together, these studies show that reliable monitoring rests on three elements: clean, well-structured time-series data; performance indicators, such as efficiency and yield, that are interpreted against weather conditions; and a detection rule that maintenance staff can understand. The literature also treats gradual degradation as a normal feature of PV systems, which means that a monitoring tool must separate slow, expected ageing from acute faults.

3.8 Research Gap and Motivation

Much recent research on fault detection concentrates on complex machine-learning models. These need large amounts of labelled data, demand computing resources, and behave as black boxes that plant technicians find hard to trust. There is room for a lightweight, transparent tool that evaluates each inverter separately, places weather analysis alongside efficiency metrics in one pipeline, and detects faults with a simple efficiency threshold.

This project was motivated by that gap. It shows that descriptive analytics with standard Python libraries can provide inverter-level visibility, weather-aware interpretation and clear fault flags without predictive modelling.

3.9 Chapter Summary

This chapter reviewed the literature on solar plant monitoring, data analytics, inverter efficiency, weather effects and fault detection. It showed that monitoring has moved from manual reading to automated data acquisition, that performance indicators must be interpreted against weather, and that fault detection ranges from simple thresholds to complex machine-learning models. A

table of selected studies was presented, and the review identified a gap for a transparent, inverter-level, weather-aware tool based on descriptive analytics, which motivates the present project.

CHAPTER 4: SOLAR POWER SYSTEM FUNDAMENTALS

4.1 The Photovoltaic Effect

Solar energy is the radiant light and heat produced by nuclear fusion in the sun. The earth intercepts far more of it in a short time than humanity consumes in a year, but the challenge lies in converting it into usable electricity. Photovoltaic technology does this directly, without turbines or moving parts, whereas solar thermal systems first convert the energy into heat. The advantages of PV are a clean and renewable resource, zero emissions in operation, low running costs after installation and flexible deployment from a single rooftop to a utility-scale plant.

A solar panel works through the photovoltaic effect: the generation of voltage and current in a material exposed to light. The effect was first observed by Edmond Becquerel in 1839 [16], and practical solar cells followed only with the development of semiconductor physics in the twentieth century.

A cell is made of silicon whose two layers are doped with different impurities, forming a P-N junction with a built-in electric field. When sunlight strikes the cell, photons with sufficient energy knock electrons loose from their atomic bonds. The built-in field pushes these electrons in one direction, and metal contacts collect them into an external circuit as direct current (Figure 4.1). The panel does not store sunlight; it converts it continuously while light is present.

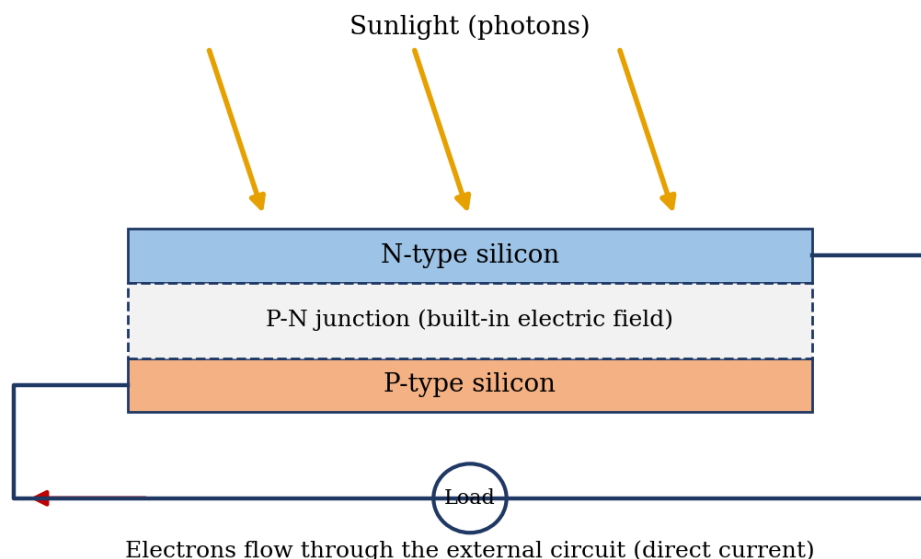


Figure 4.1: Photovoltaic Effect in a Solar Cell

4.2 Components of a Solar Power Plant

A commercial plant is an integrated electrical system, and understanding each part is necessary to interpret the data it produces. Figure 4.2 shows the flow of energy from the panels through the inverter to the grid, with sensors and monitoring devices recording data along the way. Table 4.1 summarises the role of each component.

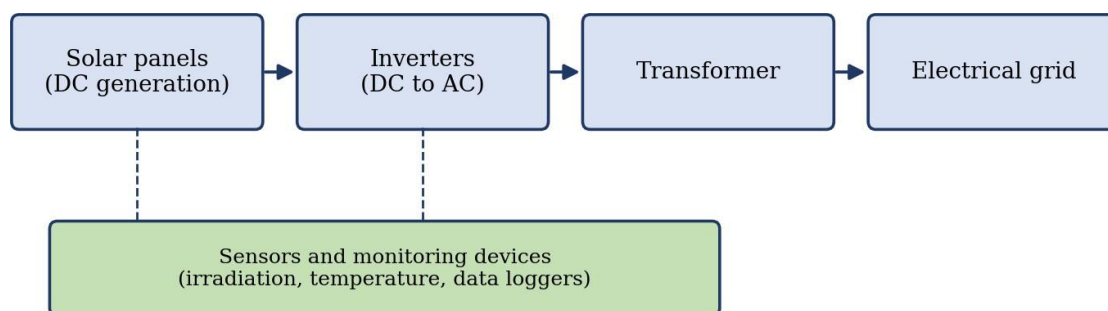


Figure 4.2: Components of a Solar Power Plant Table

4.1: Major Components of a Solar Power Plant

Component	Function	Importance
Solar panels	Absorb sunlight and convert it into DC electricity.	Source of all generation; sets the maximum yield.
Inverters	Convert DC electricity into grid-compatible AC.	Make the power usable; main focus of this project.
Sensors	Measure irradiation and temperature.	Provide the context needed to judge performance.
Monitoring devices	Collect and transmit electrical and sensor data.	Enable analytics and automated fault detection.

Solar panels (PV modules) are the most visible part of the plant. Each is a set of interconnected silicon cells that carry out the photovoltaic effect. Monocrystalline panels are cut from a single continuous crystal, which makes them more efficient and space-effective but slightly costlier, while polycrystalline panels are cheaper and somewhat less efficient. Panels are the sole source of energy in the plant, so any obstruction, degradation or damage directly reduces the potential yield.

Inverters are the most critical component from an operational and monitoring point of view. They act as the electrical gateway between the array and the grid, using solid-state switching circuits to shape DC into a smooth AC sine wave that matches the grid's voltage and frequency (50 Hz in India). No inverter is perfectly efficient, since a small share of power is lost as heat, and a high-quality inverter keeps this loss low. Because an inverter sits at the convergence point

of many panels, its failure takes the whole connected section of the plant offline, which is why inverter performance is a core module of this project.

Sensors and monitoring devices produce the data needed to judge plant performance. Pyranometers measure irradiation in W/m^2 , and temperature probes record both the ambient air temperature and the temperature of the module surface. Data loggers poll the sensors and the inverters (DC input, AC output, voltage and current) and store the readings, commonly as CSV files or databases. They form the nervous system of a plant that is managed through data.

4.3 Solar Power Generation Process

Electricity reaches the grid through five sequential steps: (1) sunlight strikes the panels; (2) photovoltaic conversion excites electrons in the cells; (3) the electrons are collected as DC and carried by cables from series-connected strings; (4) the inverter converts DC to AC and synchronises it with the grid; and (5) a step-up transformer raises the voltage for delivery to the grid. Each step can be checked through the data, which is why the analysis in this project compares DC input with AC output.

4.4 Factors Affecting Solar Power Generation

Although the theoretical capacity of a plant is set by its hardware, its actual daily yield is strongly influenced by external factors (Table 4.2). An operator who ignores them cannot tell a natural loss from a fault.

Solar irradiation is the density of solar power received on a surface and is the raw fuel of the plant; on clear days output rises almost linearly with it. **Temperature** has a negative effect: as module temperature rises, electrical resistance in the cells rises and efficiency falls [10], so a hot day is not necessarily a high-yield day. **Dust accumulation** (soiling) occurs when dust, sand or bird droppings collect on the glass and block photons, and heavy soiling can reduce output noticeably. **Shading** by trees, buildings, clouds or neighbouring rows is especially harmful because panels are wired in series, so shading a small part of a panel can limit the current of the whole string.

Table 4.2: Factors Affecting Solar Plant Performance

Factor	Impact on generation	Monitoring importance
Solar irradiation	Directly sets the volume of DC power generated.	Baseline for expected yield.
Temperature	Higher heat lowers cell efficiency.	Explains output loss in hot periods.
Dust accumulation	Blocks sunlight from reaching the cells.	Slow decline guides cleaning schedules.

Factor	Impact on generation	Monitoring importance
Shading	Reduces string current and voltage.	Reveals localised obstruction at specific inverters.

4.5 Economic Aspects of Solar Power

Solar plants need a high initial investment. Panels make up the bulk of the material cost, followed by inverters, mounting structures, cabling and electrical equipment, monitoring equipment and installation labour. Cost scales with capacity, generally with economies of scale for large plants, so residential systems of 3 to 10 kW cost more per unit of power than utility-scale parks.

Once installed, running costs are low because there are no moving parts. Routine work consists of cleaning the panels, checking electrical connections and mounting structures, and maintaining the inverters, which handle large electrical loads and generate heat. Good maintenance is cheap compared with the losses it prevents, and Peters and Madlener show that the choice of maintenance strategy has a measurable economic effect on PV plants [7].

The financial viability of a plant is judged by its return on investment (ROI) and its payback period:

$$ROI (\%) = (Net\ profit \div Initial\ investment) \times 100 \quad (4.1)$$

$$Payback\ period = Initial\ investment \div Annual\ savings \quad (4.2)$$

Both depend on how efficiently the plant operates, along with system cost, local irradiation and the price of grid electricity. If inverters degrade and run at 85% efficiency instead of 95%, the plant generates about a tenth less electricity than it should, and this loss reduces daily revenue and lengthens the payback period. Performance monitoring therefore protects financial returns as well as equipment.

4.6 Government Support for Solar Energy

Government support has accelerated solar adoption in India by offsetting the high initial cost. The PM Surya Ghar: Muft Bijli Yojana, announced in February 2024, aims to provide up to 300 units of free electricity per month to one crore households through rooftop solar, with subsidies paid to bank accounts and concessional bank loans [17]. Other central and state rooftop programs pay a fixed share of the installation cost, calculated on the capacity installed, which shortens the payback period for consumers.

Net metering changes the economics of decentralised solar. The system is connected to the grid through a bidirectional meter; when it produces more than the building uses, the surplus is exported and recorded, and at the end of the billing cycle the consumer pays only for net

consumption. The grid thus acts as a virtual battery. It also means that every unit lost to a faulty inverter or dirty panel is a unit that can neither be used nor credited, which is why monitoring matters for the finances of the plant as well as for its technical performance (Table 4.3).

Table 4.3: Government Support Programs for Solar Energy

Scheme	Purpose
PM Surya Ghar Yojana	Encourages rooftop solar adoption in households with subsidies and concessional loans.
Rooftop solar subsidy programs	Reduce installation cost through central and state financial assistance.
Net metering	Allows export of excess electricity and billing on net consumption.

4.7 Importance of Performance Monitoring

The fundamentals above show that a solar plant is a dynamic system. Its output depends on fluctuating weather and on power electronics that age, so continuous monitoring is an operational necessity rather than a luxury. Looking only at the total output at the end of a month is not enough. High-frequency data allows generation trends to be analysed, individual inverters to be compared and the AC/DC conversion ratio to be tracked, so that slight degradation in inverter circuitry becomes visible before it grows.

Monitoring is most useful when it includes weather. By aligning irradiation and temperature data with power generation logs, operators can judge whether a drop in output is explained by conditions or points to a fault. This contextualisation is what makes accurate fault detection possible, and it is the reason the analytical modules of this project are combined in a single pipeline.

4.8 Performance Indicators

Standards such as IEC 61724-1 define indicators for describing the performance of a PV system [1]. The final yield is the AC energy delivered per unit of installed DC capacity, the reference yield is the energy that the available irradiation could deliver at rated conditions, and the performance ratio is their quotient:

$$\text{Final yield } Y_f = E_{AC} \div P_0 \text{ (kWh/kWp)} \quad (4.3)$$

$$\text{Reference yield } Y_r = H \div G_{STC} \quad (4.4)$$

$$\text{Performance ratio } PR = Y_f \div Y_r \quad (4.5)$$

Here E_{AC} is the AC energy over a period, P_0 the rated DC capacity, H the irradiation on the array plane over the period and G_{STC} the reference irradiance of 1000 W/m². Because the

performance ratio depends on weather as well as on system quality, weather-corrected versions have been proposed [4]. The dataset used in this project does not include the rated capacity of the plant, so the performance ratio could not be computed. The project therefore relies on two indicators that the data does support: cumulative yield per inverter and the AC/DC conversion efficiency, both interpreted alongside the weather variables.

4.9 Chapter Summary

This chapter explained the photovoltaic effect, the components of a solar plant, the sequential generation process and the external factors that influence yield. It also covered the economics of solar power, the role of government schemes such as PM Surya Ghar and net metering, and the performance indicators used to describe PV systems. Together these fundamentals give the technical background for the analysis of DC power, AC power, efficiency and weather variables in the following chapters.

CHAPTER 5: DATASET AND DEVELOPMENT ENVIRONMENT

5.1 Dataset Source

The data for this project comes from the public Solar Power Generation dataset on Kaggle [18]. It was chosen because it contains real, unidealised sensor logs from grid-connected solar plants in India and provides both electrical generation data and synchronised weather data, which together allow inverter efficiency and weather effects to be calculated. It also uses inverter identifiers, so performance can be studied unit by unit rather than for the plant as a whole.

Many datasets available online are artificially generated or idealised for classroom exercises. The chosen dataset instead contains actual sensor logs recorded at the plants, with the gaps, noise and irregularities that real industrial data has, which makes it a suitable test of the cleaning and analysis methods used in this project.

5.2 Dataset Overview

The dataset is a time series in which every record is indexed by a time stamp. It has two components: generation records from the inverters, and weather records from plant-level environmental sensors. Table 5.1 gives an overview, and the two components are merged on their shared 15-minute time stamps.

Instead of one plant-wide total, the dataset identifies each inverter by a unique source key, which makes component-level performance monitoring possible. The two subsets share the same 15-minute time stamps, so each electrical reading can be matched with the weather at that moment.

Table 5.1: Dataset Overview

Dataset component	Description
Generation data	Inverter-level DC power, AC power and yield records.
Weather data	Ambient temperature, module temperature and irradiation.
Plant analysed	Plant 1
Observation period	34 days (15 May to 17 June 2020)
Sampling interval	15 minutes
Number of inverters	22

5.3 Generation Data Features

The generation data supplies the electrical baseline of the project. DC power is the direct current power produced by the panels and fed to the inverter, and AC power is what the inverter delivers; their ratio gives conversion efficiency. Daily yield is a running total of energy generated since midnight, and total yield is the cumulative lifetime energy of an inverter (Table 5.2).

Table 5.2: Generation Data Features

Column name	Description
DATE_TIME	Time stamp of the observation (15-minute interval).
PLANT_ID	Unique identifier of the solar plant.
SOURCE_KEY	Unique identifier of an individual inverter.
DC_POWER	Direct current power generated by the panels.
AC_POWER	Alternating current power output of the inverter.
DAILY_YIELD	Cumulative energy generated on the current day.
TOTAL_YIELD	Cumulative lifetime energy generated by the inverter.

5.4 Weather Data Features

The weather data explains why the plant behaves as it does. Irradiation is the primary driver of generation and is recorded on a scale from 0 to about 1.2 (kW/m²). Ambient temperature is the air temperature at the site, while module temperature is measured on the panel surface, which is what directly affects cell efficiency (Table 5.3). The source key here identifies the sensor unit rather than an inverter.

Table 5.3: Weather Data Features

Column name	Description
DATE_TIME	Time stamp of the weather observation.
SOURCE_KEY	Identifier of the environmental sensor.
AMBIENT_TEMPERATURE	Temperature of the surrounding air.
MODULE_TEMPERATURE	Surface temperature of the solar panels.
IRRADIATION	Intensity of incoming solar radiation (scale of about 0 to 1.2).

5.5 Software and Development Environment

All analysis was written in Python and executed in Google Colab, a cloud-hosted Jupyter-style notebook environment [19]. Because code runs on remote servers rather than on the user's own machine, heavy operations such as merging large datasets are not limited by local hardware.

Colab requires no installation or configuration, and standard data science libraries are already available. Its notebooks also interleave code cells with explanatory text, which gave a step-by-step record of the analysis and made the work easy to document. Spreadsheet tools were used briefly to inspect the raw CSV files.

Pandas provided the DataFrame structure for cleaning, merging and grouping [20]. NumPy supplied fast vectorised numerical operations, such as the efficiency division across thousands of rows [21]. Matplotlib generated the basic line and scatter plots [22], and Seaborn added statistical charts such as bar charts and regression plots [23]. Table 5.4 lists the tools and their roles.

Table 5.4: Software and Libraries Used

Tool	Purpose
Python	Core language for all analytics.
Google Colab	Cloud-based notebook environment for writing and running code.
Spreadsheet tools	Initial inspection of raw CSV files.
Pandas	Data manipulation, cleaning and merging.
NumPy	High-speed numerical calculations.
Matplotlib	Line charts and basic plots.
Seaborn	Statistical visualisation, bar charts and regression plots.

5.6 Data Quality Considerations

Real sensor logs contain problems that shape the analysis. Sensors occasionally lose power or connectivity, which leaves gaps or null readings; night-time records carry zero generation and must not be treated as faults; and the two files use different date-time formats, which causes misalignment if they are merged without conversion. These points explain why so much of the methodology in Chapter 6 is devoted to cleaning and preprocessing, and why the efficiency calculation is restricted to periods of positive DC power.

5.7 Chapter Summary

This chapter described the Kaggle solar power dataset, its generation and weather components, and the features in each. It also listed the software used, namely Python in Google Colab with Pandas, NumPy, Matplotlib and Seaborn. The dataset supplies both electrical and environmental data at 15-minute intervals for 22 inverters, which is what makes inverter-level efficiency and weather analysis possible.

CHAPTER 6: METHODOLOGY

6.1 Project Workflow

A structured, sequential workflow was followed so that every conclusion rests on clean, verified data (Figure 6.1). It has nine stages, listed below and described in the remainder of this chapter.

7. **Dataset collection:** acquire the raw generation and weather files for the target period.
8. **Data cleaning:** identify and correct errors, missing values and structural inconsistencies.
9. **Data preprocessing:** convert data types, standardise date-time indices and merge the datasets.
10. **Power generation analysis:** aggregate the clean data to establish a baseline of DC and AC power.
11. **Inverter performance analysis:** group data by inverter to compare individual yields.
12. **Efficiency analysis:** calculate the AC/DC ratio for each inverter as a health metric.
13. **Weather impact analysis:** correlate power and efficiency with irradiation and temperature.
14. **Fault detection:** apply rule-based thresholds to identify underperforming hardware.
15. **Result interpretation:** translate outputs and charts into maintenance insights.

The five analytical modules are described in Table 6.1.

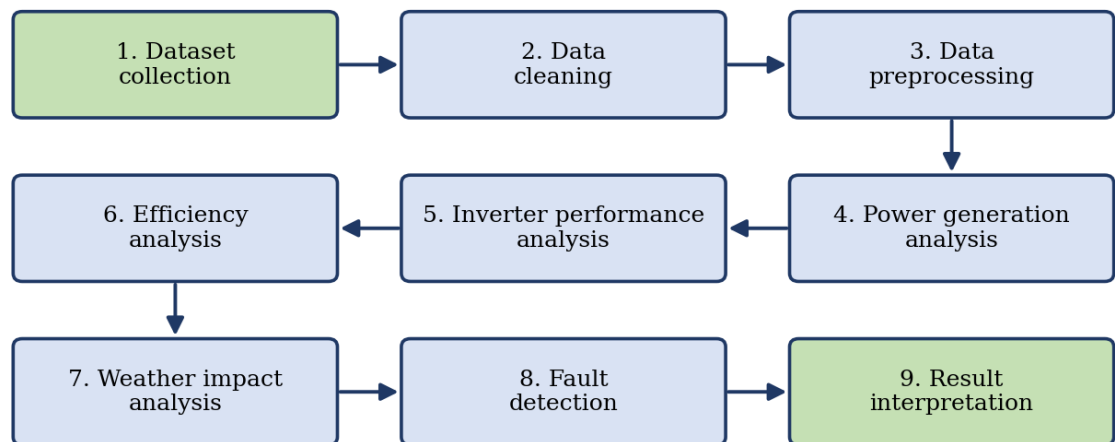


Figure 6.1: Project Workflow

Table 6.1: Analytical Modules Used in the Project

Module	Objective
Power generation analysis	Establish the baseline generation curve of the plant.
Inverter performance analysis	Compare cumulative energy yield across individual inverters.
Efficiency analysis	Calculate the AC/DC conversion ratio.
Weather impact analysis	Relate irradiation and temperature to power and efficiency.
Fault detection	Flag inverters operating below the efficiency threshold.

6.2 Data Collection

In industry, data would be pulled from the plant's SCADA system. For this project the Kaggle dataset was used instead, since it reproduces the realities of commercial monitoring without requiring access to proprietary client servers. The two CSV files, generation and weather, cover 34 days at 15-minute intervals.

6.3 Data Cleaning

Cleaning is the most time-consuming stage, because analysis of corrupted data gives false conclusions. The raw files were first inspected with Pandas functions to check their shape and statistics. Missing values were then located and either dropped or interpolated from neighbouring readings, duplicated time stamps were removed, and physically impossible values, such as negative AC power or generation at night, were flagged. Finally, time stamps stored as text were converted to Python datetime objects so that data could be sorted and grouped chronologically (Table 6.2).

Table 6.2: Data Cleaning Activities

Activity	Purpose
Missing value checking	Identify and handle null sensor readings.
Duplicate record checking	Remove repeated time stamps that inflate totals.
Consistency verification	Remove physically impossible values.
Date-time formatting	Convert text time stamps to datetime objects.

6.4 Data Preprocessing

After cleaning, the datetime column was set as the DataFrame index, turning the data into a time series that can be queried quickly by day or hour. Data types were then verified so that numeric columns such as DC power and temperature were stored as numbers rather than text, since arithmetic on text would stop the script. Finally, the generation and weather datasets were

combined with an inner merge on the shared time stamp, which aligns every 15-minute electrical reading with the weather reading taken at the same moment. Listing 6.1 shows the key steps.

Listing 6.1: Loading, time-stamp standardisation and merging (key steps)

```
import pandas as pd

gen = pd.read_csv('Plant_1_Generation_Data.csv')
wea = pd.read_csv('Plant_1_Weather_Sensor_Data.csv')

# convert text time stamps to datetime objects
gen['DATE_TIME'] = pd.to_datetime(gen['DATE_TIME'], dayfirst=True)
wea['DATE_TIME'] = pd.to_datetime(wea['DATE_TIME'])

# remove duplicates and check missing values
gen = gen.drop_duplicates()
print(gen.isnull().sum())

# inner merge on the shared time stamp
df = gen.merge(wea, on='DATE_TIME', how='inner')
df = df.set_index('DATE_TIME')
```

6.5 Exploratory Data Analysis

Exploratory analysis gave an initial understanding of the data before targeted analysis began. Simple line charts of daily yield confirmed the expected bell-shaped daily generation curve, scatter plots helped reveal anomalies that cleaning had missed, and a preliminary correlation matrix confirmed that irradiation and DC power move together.

6.6 Power Generation Analysis

DC and AC readings were grouped by day and plotted on the same time axis, so that AC output could be checked to follow DC input at a slightly lower magnitude, as expected from conversion losses. Daily yield was taken as the value recorded at the end of each day (23:45), and total yield was plotted to check its steady rise.

6.7 Inverter Performance Analysis

To evaluate individual inverters, the data was grouped by the SOURCE_KEY column. For each inverter the maximum total yield recorded on the final day of the observation period was extracted, and these values were plotted side by side in a Seaborn bar chart. This separates high-performing units from those whose cumulative yield lags behind, which points to units needing inspection.

6.8 Efficiency Analysis

Yield comparison shows which inverters produce less but not why. A new column was therefore created holding conversion efficiency for every 15-minute record:

$$\eta (\%) = (AC \text{ power} \div DC \text{ power}) \times 100 \quad (6.1)$$

The calculation was applied only where DC power was strictly greater than zero. In Plant 1's file the DC column is reported on a scale ten times larger than the AC column, so DC values were divided by 10 before the ratio was taken. Without the positive-power condition, the night-time zeros would produce undefined values (0/0) and distort the results. Efficiency is a useful indicator because it removes environmental effects: a shaded inverter yields less energy, but its conversion efficiency should stay high, so a genuine fall in efficiency points to hardware degradation.

6.9 Weather Impact Analysis

A weather-aware analysis prevents natural variation from being misread as a fault. AC power was plotted against irradiation in a scatter plot to verify that the arrays respond linearly to sunlight. Ambient and module temperature are carried in the merged dataset as additional context variables, since heat is known to lower the efficiency of silicon modules [10]. The resulting relationships form a weather-contextualised baseline against which deviations can be judged.

6.10 Fault Detection Methodology

Because the project scope excludes machine learning, fault detection uses a transparent efficiency rule. An inverter is flagged when its conversion efficiency drops below 90% during daylight hours, a threshold chosen to be below normal inverter efficiency so that minor fluctuations are ignored. The rule was implemented with Pandas conditional filtering, which scans each inverter's efficiency and returns the flagged time stamps and SOURCE_KEY values. When DC input is high but AC output is abnormally low, the ratio falls, the threshold is breached and the fault is reported.

Listing 6.2 shows how the efficiency column and the fault rule were implemented.

Listing 6.2: Efficiency calculation and 90% threshold rule (key steps)

```
# efficiency only where DC power is greater than zero
df = df[df['DC_POWER'] > 0].copy()
df['EFFICIENCY'] = df['AC_POWER'] / (df['DC_POWER'] / 10) * 100

# flag daylight records below 90% efficiency
faults = df[df['EFFICIENCY'] < 90]
print(faults.groupby('SOURCE_KEY').size())
```

Table 6.3 collects the main parameters and rules used across the analytical modules.

Table 6.3: Key Parameters and Rules Used in the Analysis

Parameter	Value or rule
Sampling interval	15 minutes
Merge type	Inner merge on DATE_TIME
Daily yield	Value recorded at 23:45 on each day
Lifetime yield per inverter	Maximum TOTAL_YIELD per SOURCE_KEY on the final day
Efficiency	$(AC_POWER \div (DC_POWER \div 10)) \times 100$, only where $DC_POWER > 0$
Fault rule	Efficiency below 90% during daylight hours

6.11 Verification of Results

Each result was checked against physical expectations before being accepted. AC power was confirmed to follow DC power at a slightly lower level, generation was confirmed to be zero at night and to peak around midday, and efficiency values were confirmed to stay below 100%. Drops in yield were cross-referenced with the irradiation and temperature records so that weather-driven dips were not reported as faults. Results that violated these expectations were traced back to cleaning or merging steps and corrected.

6.12 Chapter Summary

This chapter presented the workflow followed in the project: data collection, cleaning and preprocessing, followed by power generation, inverter performance, efficiency, weather impact and fault detection modules. Efficiency was defined as the AC/DC ratio computed only for positive DC power, and faults were defined as daylight efficiency below 90%. The key parameters are collected in Table 6.3 and the main steps are shown in Listings 6.1 and 6.2.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Introduction

This chapter presents the results of the analysis of Plant 1, which has 22 inverters and was observed for 34 days at 15-minute intervals (15 May to 17 June 2020). The methodology explains how the data was processed; this chapter explains what the processing revealed. The figures are outputs of the Python notebook described in Chapter 6, and each is discussed in terms of what it means for plant operation and maintenance. The order follows the workflow: plant-level generation, inverter-level comparison, efficiency, weather effects and finally fault detection.

7.2 Power Generation Analysis Results

Figure 7.1 plots the daily total DC and AC power of the plant across the observation period, with the DC series scaled to the AC axis for comparison. Both curves rise and fall together in a wave-like pattern. Daily totals range from roughly 470,000 on the weakest days to nearly 790,000 on the best days, and the weak days are followed by recovery within a day or two. The AC curve stays consistently just below the DC curve, which shows that the conversion loss is small and steady.

The operational significance of this graph lies in the baseline it sets. If the daily yield had dropped and not recovered, this would indicate a plant-wide failure. Because the dips recover, the plant infrastructure is sound and the dips are most likely due to weather, which is confirmed by the irradiation analysis in Section 7.5.

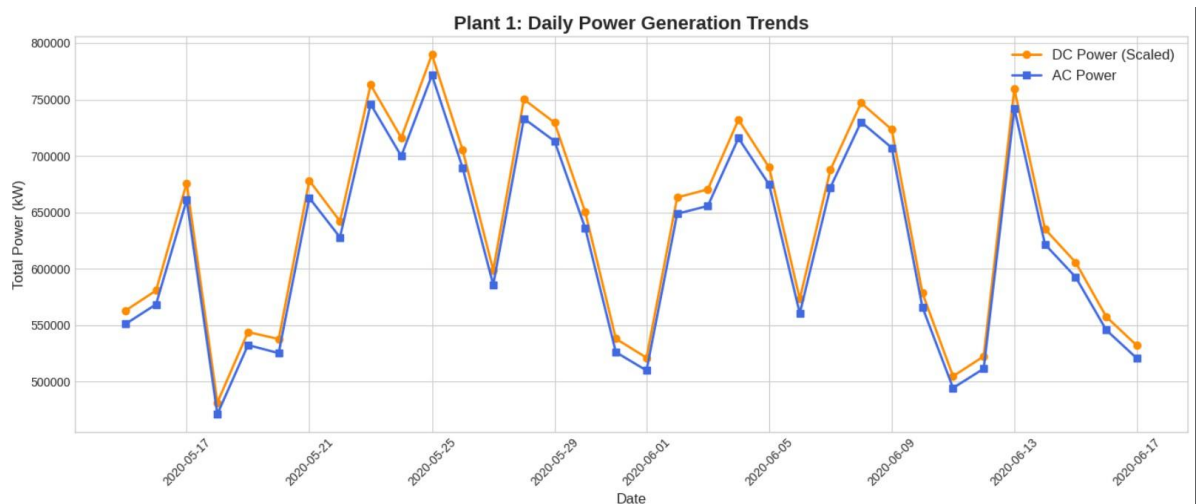


Figure 7.1: Daily Power Generation Trends

7.3 Inverter Performance Analysis Results

The cumulative (lifetime) yield of all 22 inverters is compared in Figure 7.2. The highest value is about 7.85 million kWh and the lowest about 6.43 million kWh. Yields decline gradually along the ranking rather than forming a flat baseline with a few sharp outliers, and most inverters lie between about 6.5 and 7.5 million kWh.

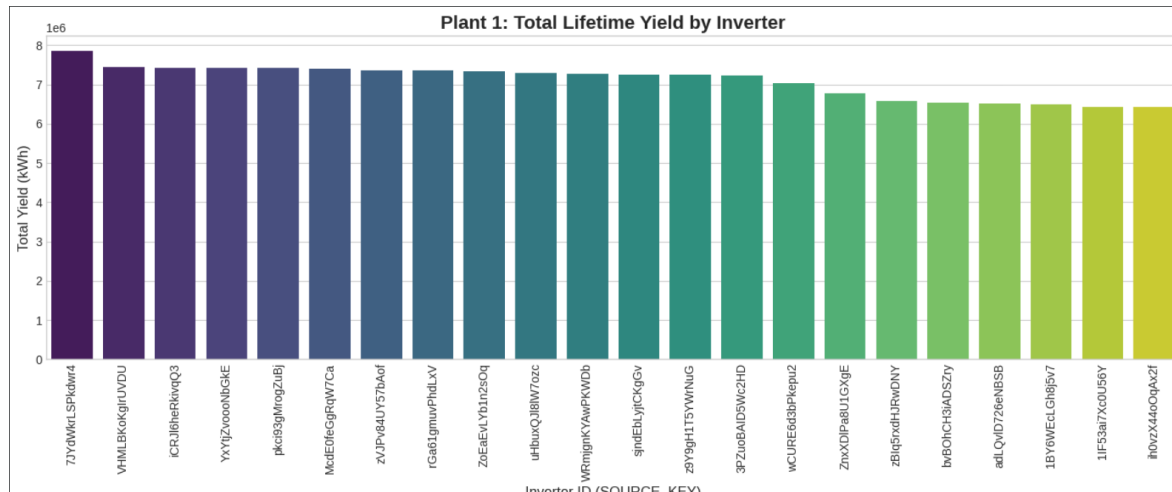


Figure 7.2: Total Lifetime Yield by Inverter

Figure 7.3 lists the ten best inverters, whose yields lie between about 7.29 and 7.85 million kWh. The leading unit is clearly ahead of the second (about 7.46 million kWh), and these units form the benchmark for the plant, since all inverters share the same site and climate.

	INVERTER_ID	TOTAL_YIELD
0	7JYdWkrLSPkdwr4	7846821.0
1	VHMLBKoKglrUVDU	7456208.0
2	iCRJI6heRkivqQ3	7426263.0
3	YxYtjZvoooNbGkE	7425442.0
4	pkci93gMrogZuBj	7415430.0
5	McdE0feGgRqW7Ca	7408587.0
6	zVJPv84UY57bAof	7363272.0
7	rGa61gmuvPhdLxV	7356897.0
8	ZoEaEvLYb1n2sOq	7341753.0
9	uHbuxQJl8IW7ozc	7287002.0

Figure 7.3: Top 10 Performing Inverters

Figure 7.4 lists the five lowest inverters, with yields between about 6.43 and 6.54 million kWh, which is 17% to 18% below the leader. Since neighbouring inverters at the same site perform better, weather cannot explain the gap. Two points affect its interpretation. The total yield is a

lifetime figure, so part of the gap may reflect the history of each unit rather than its recent behaviour. And, as Section 7.4 shows, the conversion efficiency of the plant is healthy, so the shortfall is more likely to come from the DC side (soiling, shading or string differences) than from inverter faults. These five units are the first candidates for inspection.

Bottom 5 Performing Inverters:

	INVERTER_ID	TOTAL_YIELD
17	bvBOhCH3iADSZry	6539009.0
18	adLQvID726eNBSB	6524508.0
19	1BY6WEcLGh8j5v7	6485319.0
20	1IF53ai7Xc0U56Y	6433566.0
21	ih0vzX44oOqAx2f	6426129.0

Figure 7.4: Bottom 5 Performing Inverters

This comparison shows the value of inverter-level analytics. If a manager looked only at total plant output, the lower yield of a few units would be hidden by the output of the others. By ranking the inverters, the analysis points maintenance staff to specific units and supports targeted, cost-effective inspection instead of a check of the whole plant.

7.4 Efficiency Analysis Results

Figure 7.5 shows a sample of the calculated efficiency data. For the five inverters shown at 06:00 on 15 May 2020, efficiency lies between 96.3% and 96.6%, so the ratio is already consistent between units in the early-morning low-power period.

Sample of calculated efficiency data:

	DATE_TIME	INVERTER_ID	DC_POWER	AC_POWER	EFFICIENCY (%)
510	2020-05-15 06:00:00	1BY6WEcLGh8j5v7	37.142857	3.585714	96.538462
511	2020-05-15 06:00:00	1IF53ai7Xc0U56Y	53.500000	5.162500	96.495327
512	2020-05-15 06:00:00	3PZuoBAID5Wc2HD	58.000000	5.585714	96.305419
513	2020-05-15 06:00:00	7JYdWkrLSPkdwr4	58.428571	5.628571	96.332518
514	2020-05-15 06:00:00	McdE0feGgRqW7Ca	54.375000	5.250000	96.551724

Figure 7.5: Sample of Calculated Efficiency Data

Figure 7.6 plots the average daily conversion efficiency of the plant against the 90% fault threshold. The efficiency stays between roughly 97.5% and 98% throughout the 34 days, with only very small fluctuations and no downward trend, and it remains about eight percentage points above the threshold line. This stability shows that the internal electronics of the inverters are healthy and that the conversion loss is small.

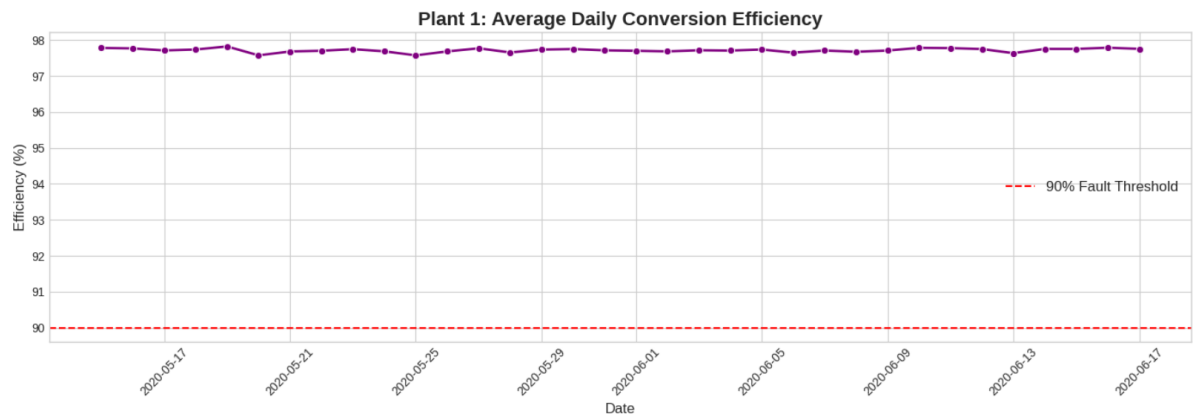


Figure 7.6: Average Daily Conversion Efficiency

7.5 Weather Impact Analysis Results

Figure 7.7 plots AC power against solar irradiation. The points form a dense band that rises almost linearly with irradiation, reaching about 1,400 kW at irradiation values above 1.0. Irradiation is therefore the main driver of plant output, and the day-to-day dips in Figure 7.1 are consistent with days of lower irradiation, that is, with cloud cover rather than hardware faults. The scatter within the band comes from differences between inverters and from short-term cloud effects. A small horizontal line of points at zero AC power, even at substantial irradiation, marks intervals in which an inverter was not generating despite available sunlight, for example during outages or shutdowns. These intervals are few compared with the whole dataset but deserve a check in the plant's event logs. The negative effect of module temperature on silicon efficiency is well documented [10] and remains a useful variable for extending this analysis.

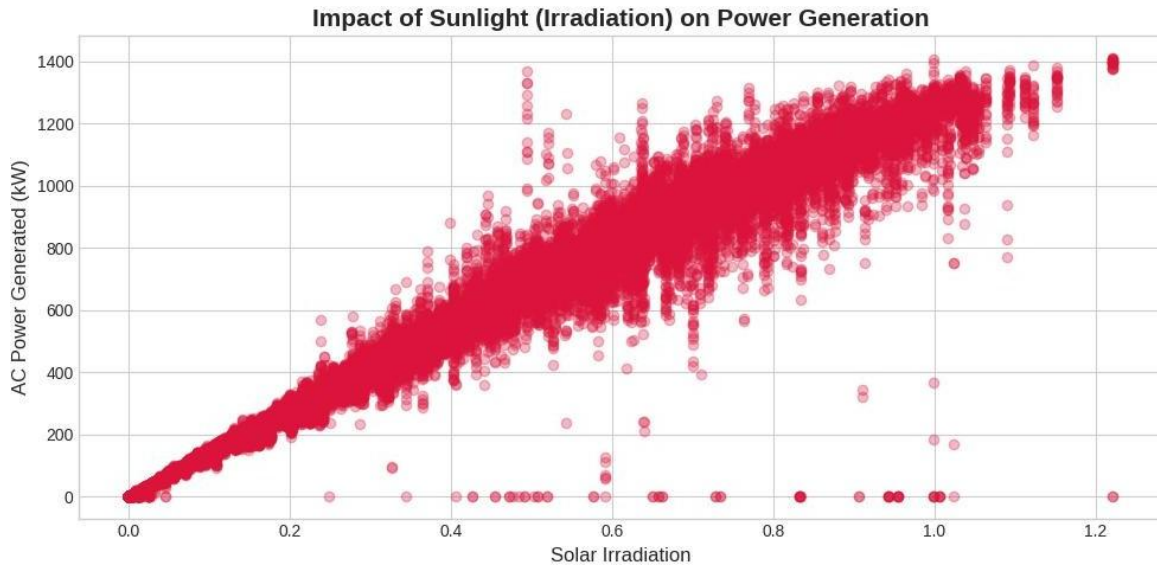


Figure 7.7: Impact of Irradiation on Power Generation

7.6 Fault Detection Results

Figure 7.8 shows the number of efficiency faults, meaning daylight records with efficiency below 90%, recorded for each inverter. The chart is empty: no inverter breached the threshold in any daylight interval. This agrees with the stable efficiency of about 97.5% to 98% in Figure 7.6 and confirms that none of the 22 inverters showed a conversion fault during the period. The rule therefore raised no false alarms on healthy data. Because the dataset contains no faulty inverter, the rule has not yet been tested against a real fault, which is a limit of this study.

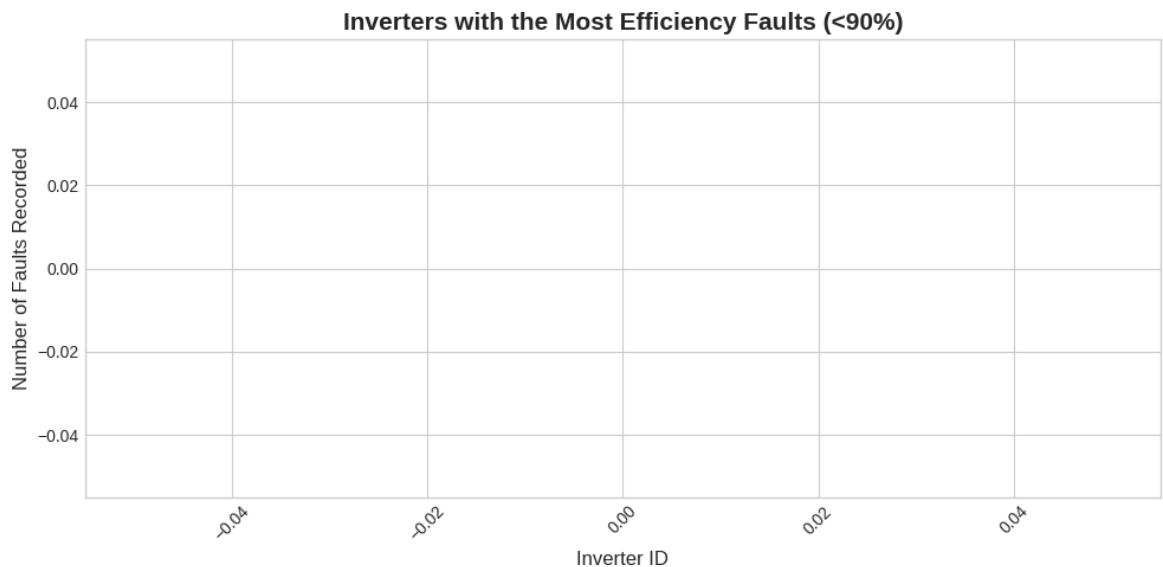


Figure 7.8: Inverters with Efficiency Faults Below 90%

For plant operators, an empty fault chart is itself useful information. It shows that the inverters are converting power as expected and that the low yield of the bottom five units should be

investigated on the panel side. The same rule can be run on new data every day, so that a future efficiency drop would be reported with the exact inverter and time stamp.

7.7 Overall Findings

Table 7.1 consolidates the findings. Together they show that the plant follows a healthy daily generation cycle, that lifetime yields differ by up to about 18% between the best and worst inverters, that conversion efficiency is stable at about 97.5% to 98%, that irradiation dominates output, and that no inverter fell below the 90% efficiency threshold.

Table 7.1: Summary of Major Findings

Analysis module	Key observation
Power generation	Daily DC and AC totals rise and fall together; AC stays just below DC.
Inverter performance	Lifetime yields range from about 6.43 to 7.85 million kWh; the five lowest are 17% to 18% below the best.
Efficiency analysis	Average daily efficiency is stable at about 97.5% to 98%, well above 90%.
Weather impact	AC power rises almost linearly with irradiation; yield dips coincide with weaker sunlight.
Fault detection	No inverter recorded efficiency below 90% in daylight hours.

7.8 Practical Significance and Limitations

The results have practical value for the management of renewable energy assets. Historically, solar plants were maintained reactively, and a broken component was noticed only when the monthly energy bill was far lower than expected. Inverter-level analysis changes this: by identifying the SOURCE_KEY of a lower-yielding unit and checking its efficiency against the plant average, managers can direct inspection to the right place before losses grow. Table 7.2 summarises how the observed patterns can be read and what action each one suggests.

Table 7.2: Interpretation of Observed Patterns

Observed pattern	Likely cause	Suggested action
Plant-wide dip in daily yield with low irradiation	Cloud cover or other weather	No maintenance needed; continue monitoring
One inverter low while neighbours are normal	Local soiling, shading or degradation	Inspect and clean connected strings
Efficiency below 90% in daylight	Inverter conversion loss	Schedule inverter service
Zero AC power despite irradiation	Outage or shutdown of an inverter	Check event logs and restart records

Observed pattern	Likely cause	Suggested action
Gradual divergence from plant average	Slow degradation of a unit	Track trend; plan repair before failure

The study also has limitations. It is based on a single 34-day dataset from one plant, so seasonal effects are not captured. Lifetime yield includes the history of each inverter, the interpretation of the charts is descriptive rather than statistical, and the 90% threshold is a fixed engineering rule rather than one learned from data. These limits define the boundary of what the analysis can claim and point to the extensions listed in Chapter 8.

7.9 Discussion in Relation to Earlier Studies

The findings are consistent with the literature reviewed in Chapter 3. The near-linear link between irradiation and power supports the view that output must be read against weather [4]. The stable, high efficiency agrees with the high conversion efficiency expected of modern inverters, and the empty fault chart shows that a simple efficiency threshold can avoid false alarms, which supports the rule-based approach that the reviews describe [11], [12]. Unlike much recent work, the project reaches these conclusions without machine learning, which keeps the method transparent and easy for plant staff to reproduce.

7.10 Chapter Summary

This chapter presented the results for Plant 1. Daily DC and AC power rise and fall together with small conversion loss; lifetime yields of the 22 inverters range from about 6.43 to 7.85 million kWh, with the five lowest units 17% to 18% below the best; conversion efficiency stays at about 97.5% to 98%; AC power rises almost linearly with irradiation; and no inverter fell below the 90% efficiency threshold. The findings were interpreted for maintenance use, and the limits of the study were stated.

CHAPTER 8: CHALLENGES, LEARNING OUTCOMES, CONCLUSION AND FUTURE SCOPE

8.1 Challenges Faced

Working with real sensor data raised several practical difficulties. The volume of 15-minute records and unfamiliar terms such as DC power and module temperature took time to understand. Missing values caused efficiency calculations to fail, and differently formatted datetime columns caused misalignment when merging the files. Separating hardware faults from weather-driven drops and designing a rule sensitive enough to catch failing hardware yet robust to noise also required care. Table 8.1 lists the challenges and the solutions applied.

Table 8.1: Challenges and Solutions

Challenge	Solution
Understanding a large raw dataset	Exploratory analysis to map the variables.
Script crashes from missing data	Pandas functions to handle NaN values.
Merge failures between files	Standardised datetime objects before merging.
Distinguishing faults from weather	Cross-referenced yield drops with irradiation and temperature.
Detecting faults without machine learning	Static efficiency threshold at 90%.

8.2 Learning Outcomes

The internship converted classroom knowledge into working skills. The main outcomes are:

- **Solar energy concepts:** practical knowledge of the photovoltaic effect, the need for DC-to-AC inversion, and the influence of irradiation and temperature on silicon panels.
- **Industrial data handling:** ability to clean and structure messy, incomplete sensor logs into a reliable dataset.
- **Python programming:** fluency in Pandas for cleaning, merging and grouping, NumPy for vectorised calculations such as efficiency across thousands of rows, and Matplotlib and Seaborn for visualisation.
- **Analytical thinking and problem solving:** treating a large dataset as a problem to be solved by logical deduction, and debugging errors and data-type conflicts independently.

- **Documentation and planning:** writing commented code and a formal report, and breaking a large objective such as solar monitoring into sequential tasks.

8.3 Conclusion

The project met its objectives. Using descriptive analytics in Python, it cleaned and merged raw sensor data, verified the healthy daily generation cycle of Plant 1, ranked all 22 inverters by lifetime yield and identified the five lowest, which are 17% to 18% below the best unit, and showed through the AC/DC ratio that conversion efficiency stayed at about 97.5% to 98%, well above the 90% threshold. The weather analysis confirmed the strong positive relationship between irradiation and AC power, which prevents natural variation from being mistaken for a fault.

A simple threshold rule flagging efficiency below 90% in daylight hours provided a transparent, lightweight fault detection tool. In this dataset it flagged no inverter, which confirms healthy conversion, and it is ready to be applied to future data. The project shows that rigorous descriptive analytics is a practical way to monitor and optimise solar infrastructure without black-box models.

Each of the objectives set in Chapter 1 was met. Data cleaning and preparation produced a reliable merged dataset; the power generation and inverter analyses gave a baseline and a ranking of the 22 inverters; the efficiency analysis provided a health metric that is independent of weather; the weather analysis contextualised the observations; and the fault detection module turned these findings into a practical maintenance tool. The internship therefore delivered both a working analytical pipeline and hands-on experience of how data analytics is used in the renewable energy industry.

8.4 Recommendations

Based on the results, the following practices are recommended for plant operators: calculate inverter efficiency daily and track it as a trend rather than as a single value; compare every inverter with the plant average so that outliers are seen early; inspect and clean the panel strings connected to the lowest-yielding inverters first; always read yield drops together with irradiation and temperature before raising a fault; and review the 90% threshold periodically against the age and rating of the inverters.

8.5 Future Scope

The current pipeline analyses static historical files. It can be extended in several practical directions, summarised in Table 8.2: forecasting of generation with regression or LSTM models,

real-time ingestion of live SCADA data, automated alerts when efficiency falls below the threshold, predictive maintenance based on degradation trends, and a web dashboard that lets non-technical plant managers view plant health.

Table 8.2: Future Enhancement Opportunities

Enhancement	Expected benefit
Machine learning forecasting	Predicts future generation from weather forecasts.
Real-time IoT monitoring	Processes live sensor data instead of static CSV files.
Automated alert systems	Sends notifications when efficiency drops below the threshold.
Predictive maintenance	Estimates degradation rates to schedule repairs before failure.
Web-based dashboards	Gives plant managers an interactive view of plant health.

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APPENDIX

```
# =====
# MODULE 1: DATA CLEANING & PREPROCESSING
# =====

import pandas as pd
import numpy as np

# 1. Load the datasets
print("Loading datasets...")
gen_df = pd.read_csv('Plant_1_Generation_Data.csv')
weather_df = pd.read_csv('Plant_1_Weather_Sensor_Data.csv')

# 2. Standardize Date-Time formats
# The generation data uses DD-MM-YYYY while weather uses YYYY-MM-DD
print("Standardizing datetime formats...")
gen_df['DATE_TIME'] = pd.to_datetime(gen_df['DATE_TIME'], format='%d-%m-%Y %H:%M')
weather_df['DATE_TIME'] = pd.to_datetime(weather_df['DATE_TIME'], format='%Y-%m-%d %H:%M:%S')

# 3. Merge datasets on Date-Time and Plant ID
print("Merging generation and weather data...")
df = pd.merge(gen_df, weather_df, on=['DATE_TIME', 'PLANT_ID'], how='inner')

# 4. Clean up column names for clarity
# SOURCE_KEY appears in both datasets.
# Gen data = Inverter ID. Weather data = Sensor ID.
df = df.rename(columns={
    'SOURCE_KEY_x': 'INVERTER_ID',
    'SOURCE_KEY_y': 'SENSOR_ID'
})

# Sort sequentially by time and inverter
df = df.sort_values(by=['DATE_TIME', 'INVERTER_ID']).reset_index(drop=True)

print("\nModule 1 Complete! Cleaned Dataset Shape:", df.shape)
display(df.head())
# =====
# MODULE 2: POWER GENERATION ANALYSIS
# =====

import matplotlib.pyplot as plt
import seaborn as sns

# Set visualization style for professional industrial charts
plt.style.use('seaborn-v0_8-whitegrid')

print("Analyzing power generation trends...")

# 1. Extract the date from the DATE_TIME column for daily aggregation
df['DATE'] = df['DATE_TIME'].dt.date

# 2. Group data by date to calculate the total daily power generation
daily_gen = df.groupby('DATE')[['DC_POWER', 'AC_POWER']].sum().reset_index()

# 3. Plot AC Power and scaled DC Power generation
fig, ax = plt.subplots(figsize=(14, 6))

# Note: DC Power in Plant 1 is scaled by 10, so we divide by 10 for accurate
visual comparison
ax.plot(daily_gen['DATE'], daily_gen['DC_POWER'] / 10, label='DC Power (Scaled)',
        color='darkorange', marker='o', linewidth=2)
ax.plot(daily_gen['DATE'], daily_gen['AC_POWER'], label='AC Power',
        color='royalblue', marker='s', linewidth=2)
```

```

ax.set_title('Plant 1: Daily Power Generation Trends', fontsize=16,
fontweight='bold')
ax.set_ylabel('Total Power (kW)', fontsize=12)
ax.set_xlabel('Date', fontsize=12)
ax.legend(fontsize=12)
plt.xticks(rotation=45)
plt.tight_layout()

print("Module 2 Complete! Displaying daily generation chart:")
plt.show()
# =====
# MODULE 3: INVERTER-WISE PERFORMANCE ANALYSIS
# =====
print("Analyzing individual inverter performance...")

# 1. Find the maximum TOTAL_YIELD recorded for each inverter
# (Since TOTAL_YIELD is cumulative, the max value represents the total lifetime
output in this dataset)
inverter_yield =
df.groupby('INVERTER_ID')['TOTAL_YIELD'].max().sort_values(ascending=False).reset
_index()

# 2. Plot the total yield comparison using a bar chart
plt.figure(figsize=(14, 6))
sns.barplot(x='INVERTER_ID', y='TOTAL_YIELD', data=inverter_yield,
palette='viridis')

plt.title('Plant 1: Total Lifetime Yield by Inverter', fontsize=16,
fontweight='bold')
plt.ylabel('Total Yield (kWh)', fontsize=12)
plt.xlabel('Inverter ID (SOURCE_KEY)', fontsize=12)
plt.xticks(rotation=90)
plt.tight_layout()

print("Module 3 Complete! Displaying inverter performance chart:")
plt.show()

# 3. Display the top and bottom performing inverters
print("\nTop 10 Performing Inverters:")
display(inverter_yield.head(10))

print("\nBottom 5 Performing Inverters:")
display(inverter_yield.tail(5))
# =====
# MODULE 4: EFFICIENCY CALCULATION
# =====
print("Calculating inverter conversion efficiency...")

# 1. Handle nighttime zero-generation to avoid division-by-zero errors
df['DC_POWER_CLEAN'] = df['DC_POWER'].replace(0, np.nan)

# 2. Calculate Efficiency: (AC Power / DC Power) * 100
# NOTE: In the Kaggle Plant 1 dataset, DC Power is scaled by a factor of 10.
# We divide by 10 to get the true efficiency percentage.
df['EFFICIENCY (%)'] = (df['AC_POWER'] / (df['DC_POWER_CLEAN'] / 10)) * 100

# 3. Calculate average daily efficiency across the plant
daily_eff = df.groupby('DATE')['EFFICIENCY (%)'].mean().reset_index()

# 4. Plot the daily average efficiency
plt.figure(figsize=(14, 5))
sns.lineplot(x='DATE', y='EFFICIENCY (%)', data=daily_eff, color='purple',
marker='o', linewidth=2)

```

```

# Add a red dashed line representing the standard 90% industry efficiency
threshold
plt.axhline(90, color='red', linestyle='--', label='90% Fault Threshold')

plt.title('Plant 1: Average Daily Conversion Efficiency', fontsize=16,
fontweight='bold')
plt.ylabel('Efficiency (%)', fontsize=12)
plt.xlabel('Date', fontsize=12)
plt.legend(fontsize=12)
plt.xticks(rotation=45)
plt.tight_layout()

print("Module 4 Complete! Displaying efficiency chart:")
plt.show()

# 5. Display a sample of the calculated data
print("\nSample of calculated efficiency data:")
display(df[['DATE_TIME', 'INVERTER_ID', 'DC_POWER', 'AC_POWER', 'EFFICIENCY
(%)']].dropna().head())
# =====
# MODULE 5: WEATHER IMPACT ANALYSIS (SIMPLIFIED)
# =====
print("Analyzing weather impact on power generation...")

# 1. Simple Text-Based Correlation (No complex heatmap)
# We calculate just the two specific relationships we care about
sunlight_corr = df['IRRADIATION'].corr(df['AC_POWER'])
temp_corr = df['MODULE_TEMPERATURE'].corr(df['EFFICIENCY (%)'])

print(f"\n--- Key Weather Insights ---")
print(f"1. Sunlight vs. Power Correlation: {sunlight_corr:.2f} (Strong
Positive)")
print(f"2. Temperature vs. Efficiency Correlation: {temp_corr:.2f}
(Negative/Weak)")
print("\n")

# 2. Single, Clear Scatter Plot: Sunlight vs Power
# This visually proves the first insight: higher irradiation increases power
output
plt.figure(figsize=(10, 5))
plt.scatter(df['IRRADIATION'], df['AC_POWER'], alpha=0.3, color='crimson')

plt.title('Impact of Sunlight (Irradiation) on Power Generation', fontsize=15,
fontweight='bold')
plt.xlabel('Solar Irradiation', fontsize=12)
plt.ylabel('AC Power Generated (kW)', fontsize=12)
plt.tight_layout()

print("Displaying simplified weather impact chart:")
plt.show()
# =====
# MODULE 6: FAULT DETECTION
# =====
print("Detecting inverter faults and performance drops...")

# 1. Filter for daytime operations (Irradiation > 0.1)
# We do this so we don't accidentally flag nighttime 0-generation as a "fault"
daytime_df = df[df['IRRADIATION'] > 0.1].copy()

# 2. Define a fault: Conversion Efficiency drops below 90%
faults_df = daytime_df[daytime_df['EFFICIENCY (%)'] < 90.0]

# 3. Count how many times each inverter had a fault
fault_counts = faults_df['INVERTER_ID'].value_counts().reset_index()
fault_counts.columns = ['INVERTER_ID', 'FAULT_COUNT']

```

```

print("\n--- Inverters Requiring Maintenance (Top 5) ---")
display(fault_counts.head())

# 4. Simple Visual: Bar chart of the inverters with the most faults
plt.figure(figsize=(10, 5))
plt.bar(fault_counts['INVERTER_ID'][:10], fault_counts['FAULT_COUNT'][:10],
color='firebrick')

plt.title('Inverters with the Most Efficiency Faults (<90%)', fontsize=15,
fontweight='bold')
plt.xlabel('Inverter ID', fontsize=12)
plt.ylabel('Number of Faults Recorded', fontsize=12)
plt.xticks(rotation=45)
plt.tight_layout()

print("\nDisplaying fault detection chart:")
plt.show()

```